

Glossary of Terms and Acronyms

Transfer Learning for Freshness Classification

Digital Image Processing — 2026

1. Acronyms and Abbreviations

2. Key Concepts

2.1. Transfer Learning

Transfer learning is a machine learning technique in which a model pre-trained on a large dataset (source domain) is adapted to perform a different but related task on a smaller dataset (target domain). In this study, all backbone models are pre-trained on ImageNet and then fine-tuned for fruit and vegetable freshness classification. The core assumption is that low-level features (edges, textures, colors) and mid-level features (shapes, patterns) learned from natural images transfer well to the food quality domain.

2.2. Backbone Architecture

A backbone is the main feature extraction component of a deep learning model for image classification. It consists of a series of convolutional layers that progressively transform a raw image into a compact feature vector. In transfer learning, the backbone is initialized with pre-trained weights and may be partially or fully frozen during training on the target task. The output of the backbone — typically obtained after global average pooling — is a fixed-length vector (e.g. 512 dimensions for ResNet-18) that summarizes the image content.

2.3. Fine-tuning

Fine-tuning refers to the process of continuing to train a pre-trained model on new data, allowing some or all of its weights to be updated. *Partial fine-tuning* (as in conditions C2 and C4) unfreezes only the last few layers of the backbone, which capture the most task-specific high-level features. *Full fine-tuning* updates all weights, which risks catastrophic forgetting unless the learning rate is carefully controlled.

2.4. Catastrophic Forgetting

A phenomenon in neural networks where learning new information causes the model to “forget” previously acquired knowledge. In transfer learning, aggressive fine-tuning of a pre-trained backbone destroys the general-purpose representations learned during pre-training, eliminating the benefit of the pre-trained initialization. This is mitigated in condition C4 by using a backbone learning rate 100 times smaller than the head learning rate ($\text{lr}_{\text{backbone}} = 10^{-5}$ vs $\text{lr} = 10^{-3}$).

2.5. Projection Head

A non-linear transformation applied between the backbone feature vector and the final linear classifier. In conditions C3 and C4, the projection head consists of two fully connected layers with ReLU activation:

$$\mathbf{f} \in \mathbb{R}^{d_{\text{out}}} \xrightarrow{\text{FC}(256)} \xrightarrow{\text{ReLU}} \xrightarrow{\text{FC}(128)} \xrightarrow{\text{ReLU}} \mathbf{h} \in \mathbb{R}^{128}$$

The projection head provides additional non-linear capacity to form more complex decision boundaries in the feature space, without requiring backbone adaptation.

2.6. Domain Shift

Domain shift refers to the statistical difference between the distribution of training data and the distribution of data encountered at test time. In this study, all backbone models are trained on internet-sourced images (typically studio photographs with controlled lighting and white backgrounds) and evaluated on real-world mobile phone photos taken under natural indoor lighting conditions. The resulting performance degradation — measured as $\text{Drop} = F1_{\text{train}} - F1_{\text{eval}}$ — quantifies the magnitude of domain shift.

2.7. Macro-Averaged Metrics

When a classification task involves multiple classes, metrics such as F1-score, precision, and recall can be computed per class and then averaged. *Macro averaging* computes the metric for each class independently and takes the unweighted mean:

$$\overline{F1} = \frac{1}{C} \sum_{c=1}^C F1_c, \quad F1_c = \frac{2TP_c}{2TP_c + FP_c + FN_c}$$

This gives equal weight to every class regardless of its sample count, which is important when minority classes (such as formalin) are particularly critical from a safety perspective.

2.8. Confusion Matrix

A square matrix of size $C \times C$ where entry (i, j) counts the number of samples of true class i predicted as class j . The diagonal entries represent correct predictions; off-diagonal entries represent misclassifications. In this study, confusion matrices are *row-normalized* so that each entry shows the fraction of samples in that true class, making error patterns directly comparable across classes of different sizes.

2.9. Cosine Annealing

A learning rate scheduling strategy in which the learning rate follows a cosine curve from its initial value down to zero over the training period:

$$\text{lr}(t) = \frac{\text{lr}_{\text{max}}}{2} \left(1 + \cos\left(\frac{\pi t}{T}\right) \right)$$

where t is the current epoch and T is the total number of epochs. This allows large gradient steps early in training and increasingly fine adjustments as the model approaches convergence.

2.10. Stratified Split

A data splitting strategy that ensures each subset (train and validation) contains the same proportion of samples from each class as the full dataset. In this study, an 80/20 stratified split is applied with a fixed random seed (42) to ensure reproducibility. This prevents scenarios where, by chance, one class is over- or under-represented in the validation set.

2.11. Formalin Treatment

Formalin (formaldehyde solution) is used in some markets to preserve the appearance of fruits and vegetables, making them visually similar to fresh produce despite being unsafe for consumption. The FruitVision dataset includes formalin-treated samples as a third class, making it the most challenging benchmark in this study. The difficulty of distinguishing formalin-treated from fresh samples in RGB images is directly analogous to the challenge of detecting subtle spoilage in fermented Korean foods.

3. Training Condition Reference

4. Dataset Reference

Acronym	Definition
AUC-ROC	<i>Area Under the Receiver Operating Characteristic Curve</i> . A metric ranging from 0.5 (random classifier) to 1.0 (perfect classifier) that measures the model’s ability to separate classes across all decision thresholds, independent of a fixed threshold.
CNN	<i>Convolutional Neural Network</i> . A class of deep learning model that applies learned filters to detect local patterns (edges, textures, shapes) in images. The foundational architecture for computer vision tasks.
C1–C4	<i>Training Conditions 1 through 4</i> . An ablation study over backbone fine-tuning and head architecture. See Section ??.
DIP	<i>Digital Image Processing</i> . The field of study concerned with the manipulation and analysis of digital images using computational methods.
EB0	<i>EfficientNet-B0</i> . A compact convolutional backbone (5.3M parameters) achieving 77.1% ImageNet top-1 accuracy through compound scaling.
EB2	<i>EfficientNet-B2</i> . A larger EfficientNet variant (9.1M parameters) achieving 80.1% ImageNet top-1 accuracy.
F1	<i>F1-Score (macro-averaged)</i> . The harmonic mean of precision and recall, averaged equally across all classes. The primary evaluation metric in this study.
FN	<i>False Negative</i> . A rotten sample incorrectly classified as fresh. The most dangerous error in food safety applications.
FP	<i>False Positive</i> . A fresh sample incorrectly classified as rotten. Results in unnecessary food waste.
GPU	<i>Graphics Processing Unit</i> . Hardware accelerator used to parallelize the matrix operations required for deep learning training and inference.
ImageNet	<i>Large Scale Visual Recognition Challenge dataset</i> . A benchmark dataset of 1.2 million images across 1,000 categories, used to pre-train backbone architectures for transfer learning.
KFQ	<i>Kaggle Fruits Quality</i> . Dataset code. 359 images, 12 mixed fruit categories, binary fresh/rotten labels.
KFR	<i>Kaggle Fruits Fresh/Rotten</i> . Dataset code. 13,599 images, apple/banana/orange, binary fresh/rotten labels.
KFS	<i>Kaggle Fresh & Stale</i> . Dataset code. 27,317 images, 9 fruit and vegetable categories, binary fresh/rotten labels.
LR	<i>Learning Rate</i> . The step size used by the optimizer when updating model weights. Controls the speed and stability of training convergence.
MCC	<i>Matthews Correlation Coefficient</i> . A balanced metric ranging from -1 (inverse prediction) through 0 (random) to $+1$ (perfect prediction). More robust than F1-score for imbalanced datasets.
MFR	<i>Mendeley Fruits Classification</i> . Dataset code. 1,655 images, peach/pomegranate/strawberry, binary fresh/rotten labels.
MFV	<i>Mendeley FruitVision</i> . Dataset code. 10,154 images, 5 fruit categories, three-class fresh/formalin/rotten labels.
MLM	<i>Mendeley Lemon Varieties</i> . Dataset code. 1,956 images, 7 lemon varieties, binary fresh/rotten labels.

ID	Backbone	Head	Full description
C1	Fully frozen	Linear only	All backbone weights locked. Single linear layer $\mathbf{W} \in \mathbb{R}^{d_{\text{out}} \times C}$ trained from scratch. Evaluates raw ImageNet feature quality.
C2	Last block free	Linear only	layer4 (final residual block) unfrozen. All earlier backbone layers frozen. Linear classifier only, no projection head. Allows high-level feature adaptation.
C3	Fully frozen	Proj. + Linear	All backbone weights locked. Two-layer projection head: $d_{\text{out}} \rightarrow 256 \rightarrow 128$, each followed by ReLU, then linear classifier. Non-linear capacity without backbone adaptation.
C4	Last block free	Proj. + Linear	layer4 unfrozen with $\text{lr}_{\text{backbone}} = 10^{-5}$. Full projection head with $\text{lr} = 10^{-3}$. Maximum adaptation capacity with catastrophic forgetting prevention.

ID	Full name	Images	Classes	Notes
KFQ	Kaggle Fruits Quality	359	fresh / rotten	12 mixed fruit types
MFR	Mendeley Fruits	1,655	fresh / rotten	Peach, pomegranate, strawberry
MLM	Mendeley Lemon Varieties	1,956	fresh / rotten	7 lemon varieties
MFV	Mendeley FruitVision	10,154	fresh / formalin / rotten	Most challenging; 3 classes
KFR	Kaggle Fruits Fresh/Rotten	13,599	fresh / rotten	Apple, banana, orange
KFS	Kaggle Fresh & Stale	27,317	fresh / rotten	9 fruit/vegetable categories
OWN	Own Dataset (strawberry)	121	fresh / rotten	Real phone photos; eval only
Total (public)		55,040		